MOHAMED SHUAIB NASURUDEEN

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EDUCATION

Worcester Polytechnic Institute **Aug. 2025 – May 2027**

Master of Science in Robotics Engineering *Worcester, MA*

- **Relevant coursework:** Motion Planning, Robot Dynamics, Robot Control, Robotics Engineering.

Puducherry Technological University **Dec. 2020 – May 2024**

Bachelor of Technology in Mechatronics Engineering, CGPA: 8.75/10 *Puducherry, India*

TECHNICAL SKILLS

- **Robot Learning:** PPO, AMP, motion priors, curricula, reward design, rollout analysis, domain randomization
- **Planning and Control:** TAMP, PDDL, Pyperplan, OMPL, trajectory optimization, IK, manipulator control
- **Simulation and Software:** MuJoCo, Isaac Lab, Genesis, ROS2, Python, C++, PyTorch, Docker, Open3D, UR RTDE
- **Hardware and Sensors:** Unitree G1, UR10e, Fanuc CRX-10iA, DOBOT, Dex3 hands, LiDAR, GigE and fisheye cameras

PROFESSIONAL EXPERIENCE

ELPIS Lab, Worcester Polytechnic Institute **Dec. 2025 – Present**

Graduate Researcher *Worcester, MA*

- Researching humanoid robot learning, whole-body control, and planning systems for locomotion, manipulation, and coordinated full-body behaviors.
- Developing MuJoCo, Isaac Lab, and Genesis pipelines for policy rollouts, task execution, debugging, and video/log-based evaluation.

10xConstruction Ai **Jan. 2025 – Jun. 2025**

Robotics Software Engineer *India*

- Developed and deployed constant-velocity and force-control behaviors on a UR10e using RTDE for construction robotics tasks.
- Integrated Fanuc CRX-10iA/L, UR10e, and Ewellix Liftkit systems with ROS2 for spraying, sanding, and manipulation workflows.
- Built multi-modal calibration scripts and ROS2 end-effector camera modules for fisheye, LiDAR, and wide-angle perception setups.

PROJECTS

Humanoid AMP-PPO: Whole-Body Policy Learning *MuJoCo, PyTorch, PPO, AMP*

- Built a Unitree G1 whole-body RL framework for locomotion and dual-arm behaviors using PPO, AMP, motion-prior rewards, curricula, and domain randomization.

Humanoid-Tugger: Whole-Body Loco-Manipulation *Isaac Lab, PyTorch, PPO*

- Developing a Unitree G1 policy for coordinated walking, torso stabilization, reaching, grasping, and cart transport with Dex3 contact rewards.

Two Towers and Beyond: Task and Motion Planning *Genesis, PDDL, Pyperplan, OMPL*

- Built a TAMP pipeline that abstracts block poses into predicates, generates PDDL problems, plans symbolic actions, executes primitives, replans, and verifies final stacks.

Learning Cost-Driven Whole-Body Planning *Motion Planning, Trajectory Optimization, LfD*

- Investigating learned motion costs from retargeted human demonstrations to guide constrained humanoid trajectory optimization with joint, collision, task, and smoothness terms.